Lab Sheet 5

IT Number: IT22548078

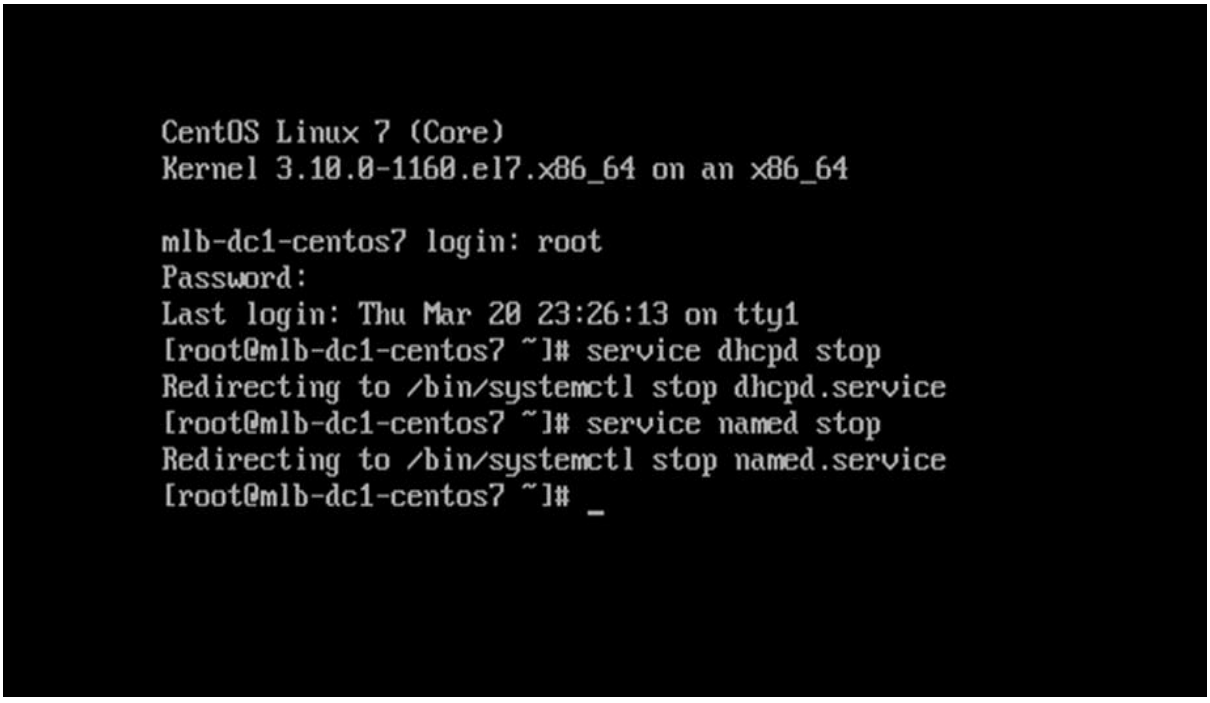
Name: M.T.K.Chathuri

Introduction

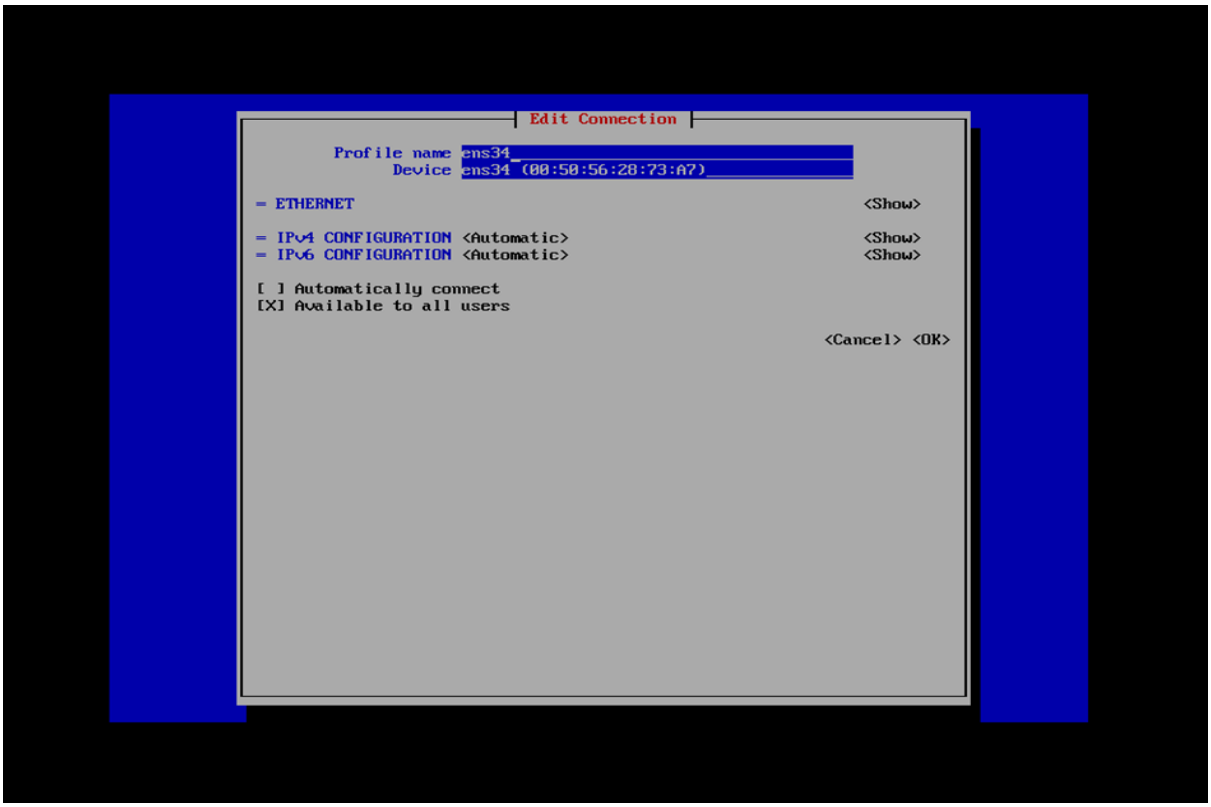
SNMP (Simple Network Management Protocol) is a protocol used for managing devices on IP networks. This report outlines the steps taken to configure SNMP on a Linux server and verify its functionality.

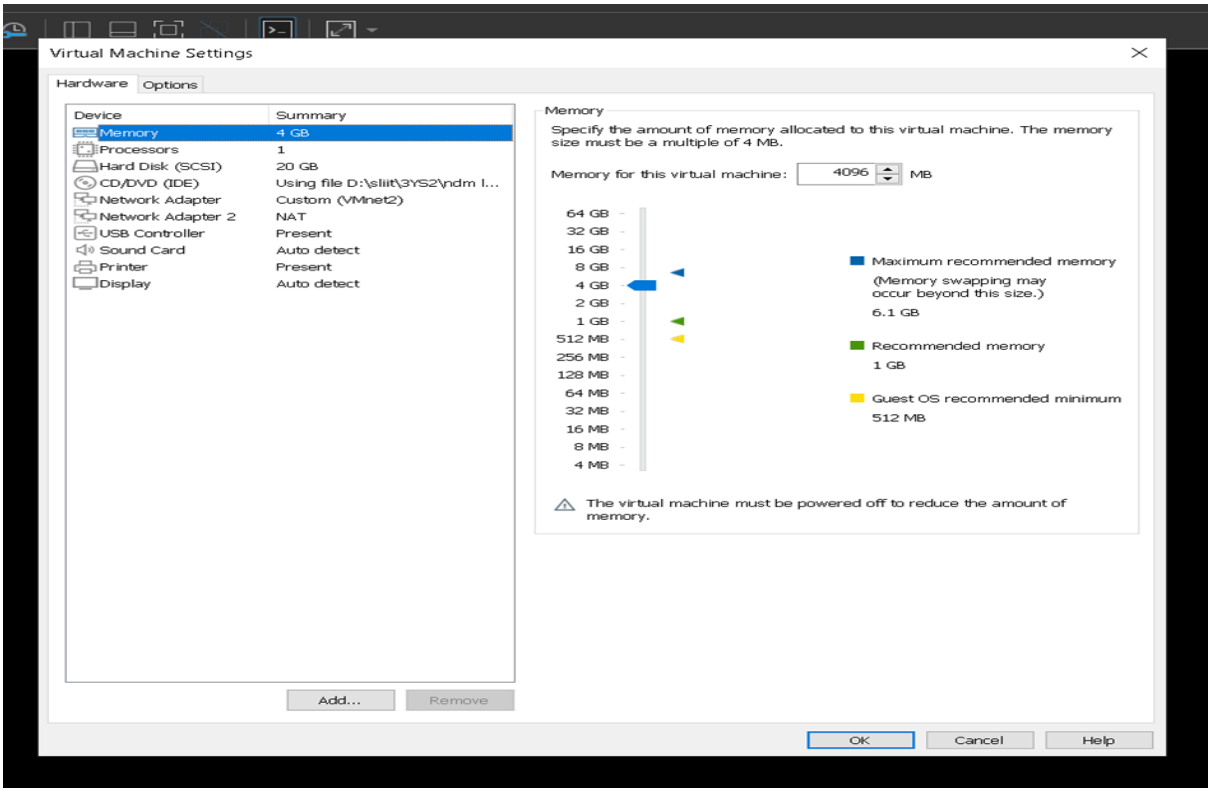
Step 1

Stop all running services.



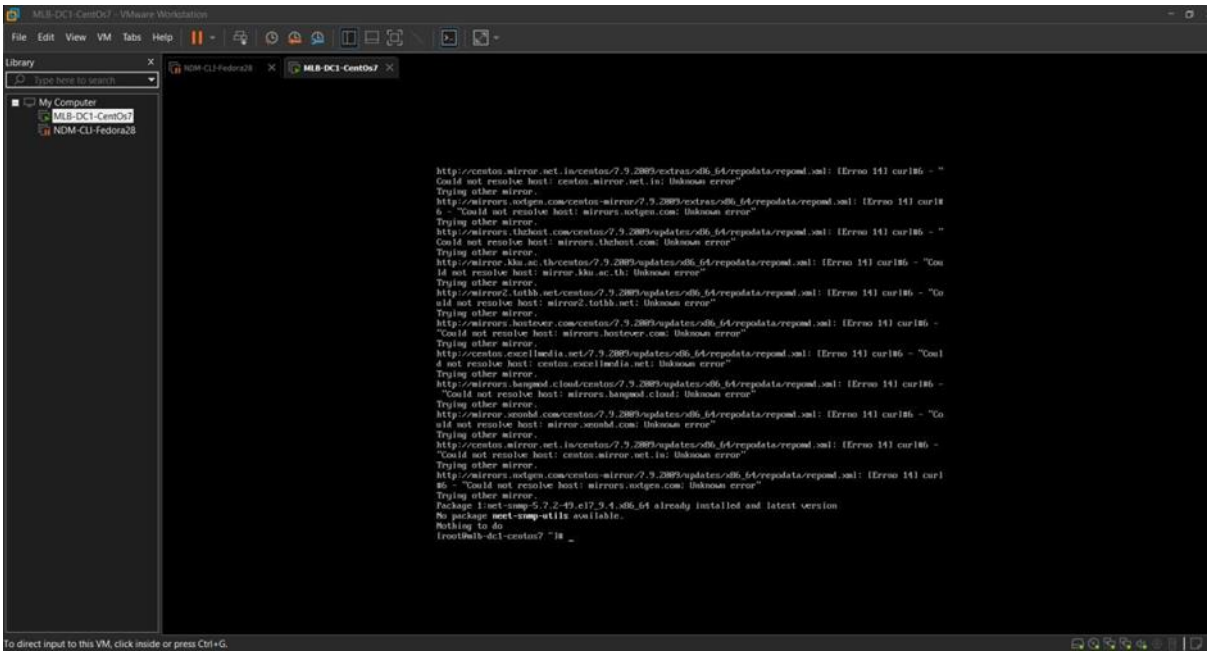
Set VMware network settings to vmnet8-NAT and make your server IP settings to obtain IP address automatically & restart the network service.





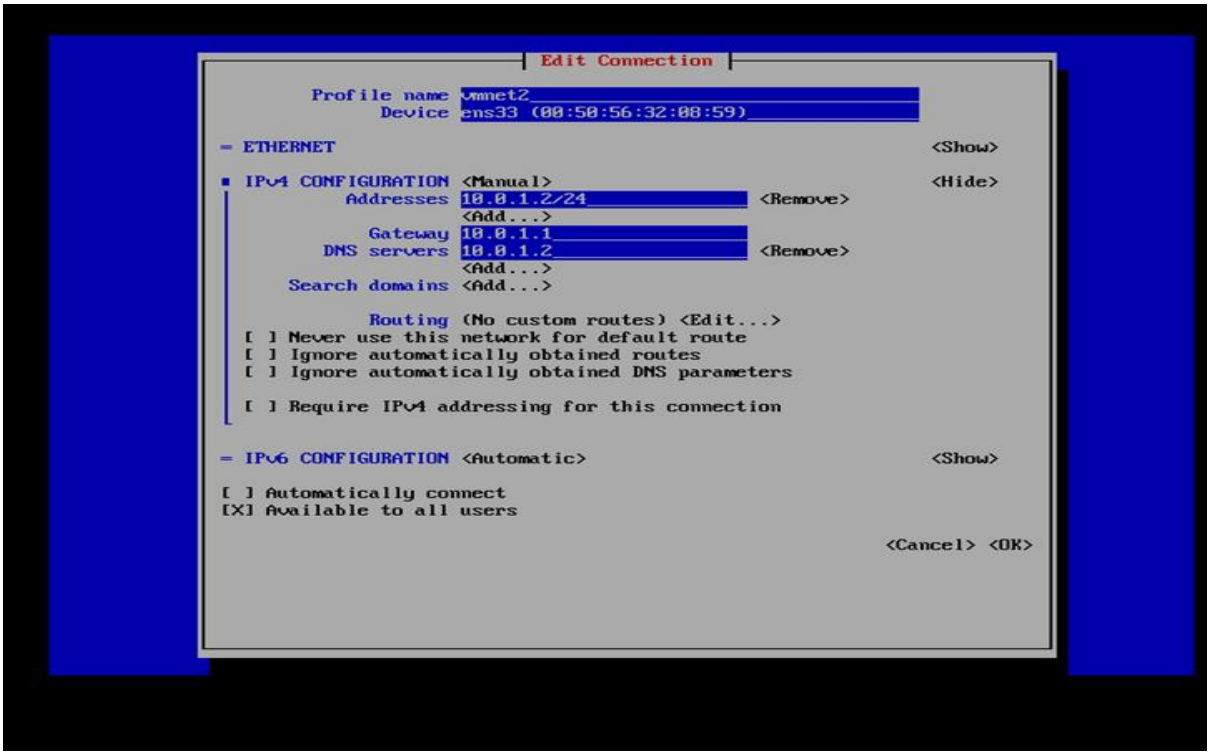
Step 2

Install SNMP using the following command.



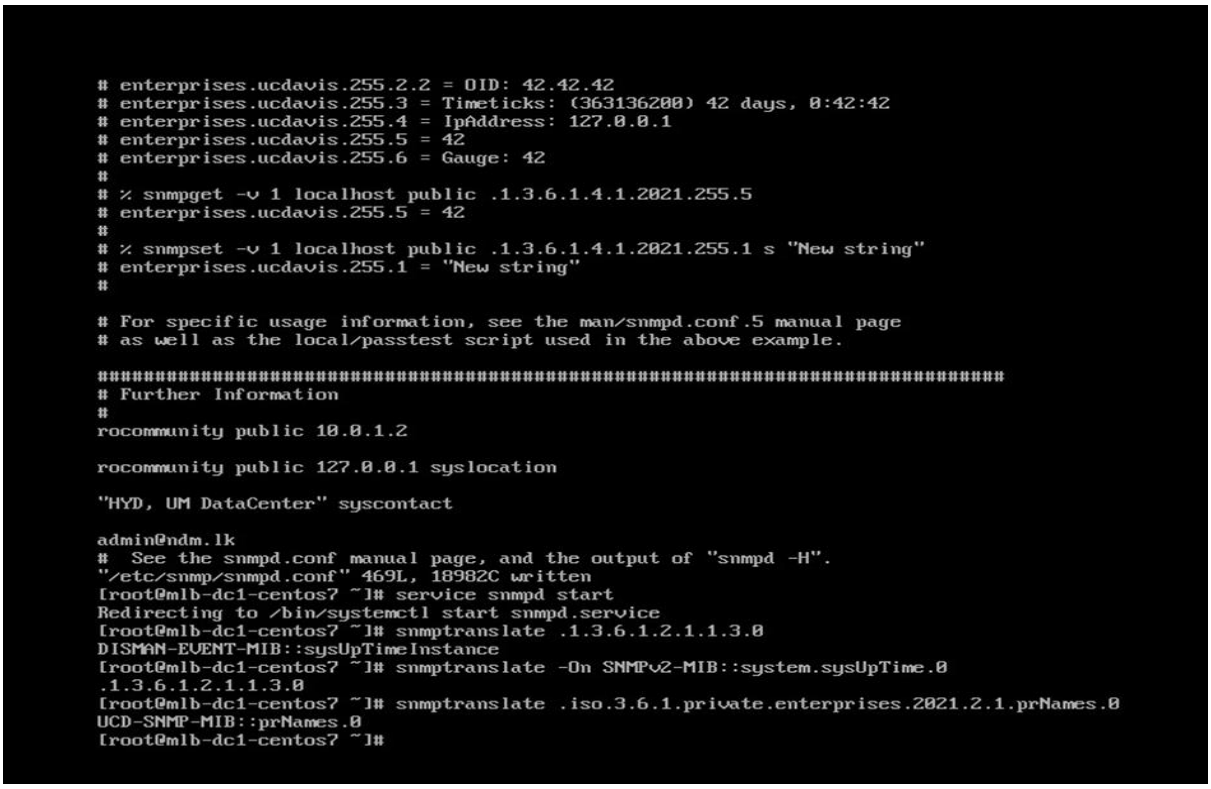
Step 3

Set VMware network settings to vmnet2-Host_only and make your server IP settings to obtain IP address manually.



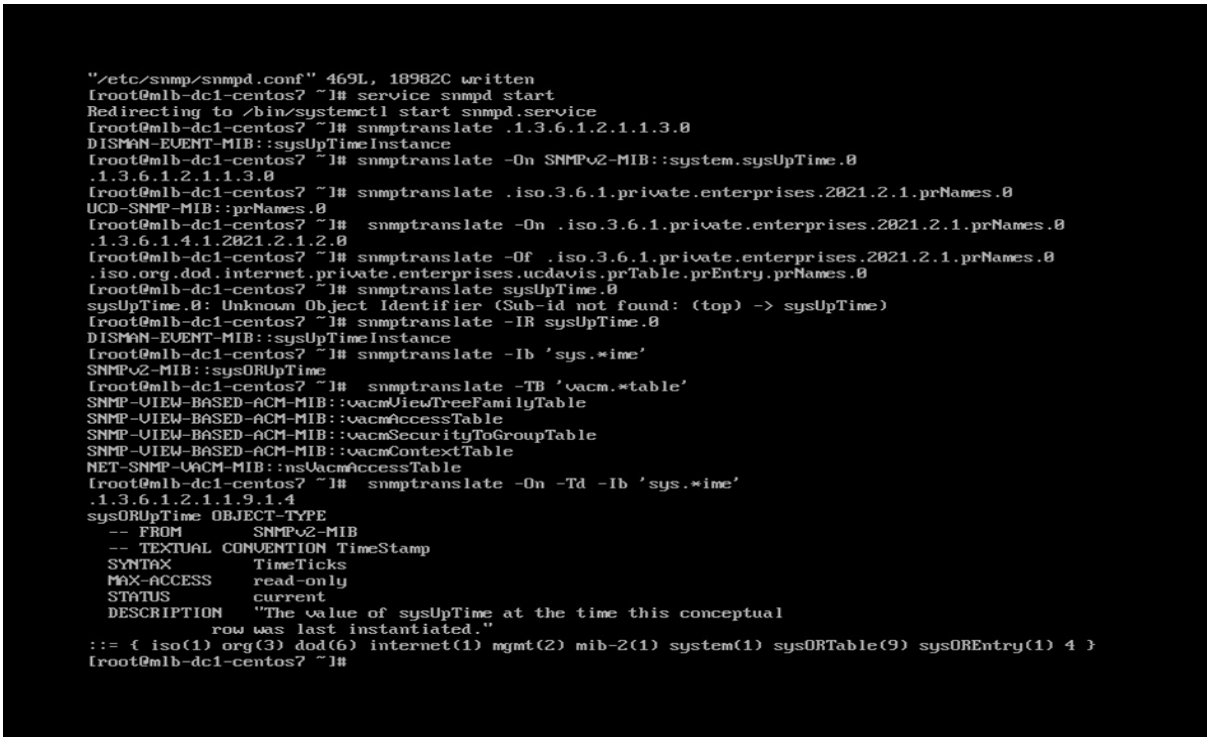
Step 4

Edit the SNMP configuration file.



Step 5

snmptranslate: learning about the MIB tree. (translate MIB OID names between numeric and textual forms)



```
        +-- -R-- String    sysDescr(1)
        |        Textual Convention: DisplayString
        |        Size: 0..255
        +-- -R-- ObjID     sysObjectID(2)
        +-- -R-- TimeTicks sysUpTime(3)
        |   |
        |   +--sysUpTimeInstance(0)
        |
        +-- -RW- String    sysContact(4)
        |        Textual Convention: DisplayString
        |        Size: 0..255
        +-- -RW- String    sysName(5)
        |        Textual Convention: DisplayString
        |        Size: 0..255
        +-- -RW- String    sysLocation(6)
        |        Textual Convention: DisplayString
        |        Size: 0..255
        +-- -R-- INTEGER   sysServices(7)
        |        Range: 0..127
        +-- -R-- TimeTicks sysORLastChange(8)
        |        Textual Convention: TimeStamp
        |
        +--sysORTable(9)
            |
            +--sysOREntry(1)
                |   Index: sysORIndex
                |
                +-- ---- INTEGER   sysORIndex(1)
                |        Range: 1..2147483647
                +-- -R-- ObjID     sysORID(2)
                +-- -R-- String    sysORDescr(3)
                |        Textual Convention: DisplayString
                |        Size: 0..255
                +-- -R-- TimeTicks sysORUpTime(4)
                         Textual Convention: TimeStamp
    [root@mlb-dc1-centos7 ~]#
    [root@mlb-dc1-centos7 ~]# _
```

snmpget: retrieving data from a host.

```
[root@mlb-dc1-centos7 ~]# snmpget -v 1 -c public 10.0.1.2 system.sysUpTime.0
DISMAN-EVENT-MIB::sysUpTimeInstance = Timeticks: (245013) 0:40:50.13
[root@mlb-dc1-centos7 ~]# snmpget -v 2c -c public 10.0.1.2 system.sysUpTime.0
DISMAN-EVENT-MIB::sysUpTimeInstance = Timeticks: (245733) 0:40:57.33
[root@mlb-dc1-centos7 ~]# snmpget -v 2c -c public 10.0.1.2 sysUpTime.0
DISMAN-EVENT-MIB::sysUpTimeInstance = Timeticks: (246534) 0:41:05.34
[root@mlb-dc1-centos7 ~]# snmpget -v 1 -c public 10.0.1.2 sysUpTime

Error in packet
Reason: (noSuchName) There is no such variable name in this MIB.
Failed object: SNMPv2-MIB::sysUpTime

[root@mlb-dc1-centos7 ~]#
[root@mlb-dc1-centos7 ~]# snmpget -v 2c -c public 10.0.1.2 sysUpTime
SNMPv2-MIB::sysUpTime = No Such Instance currently exists at this OID
[root@mlb-dc1-centos7 ~]# snmpget -v 2c -c public 10.0.1.2 sysUpTime.0

ucdDemoUserList.0DISMAN-EVENT-MIB::sysUpTimeInstance = Timeticks: (249991) 0:41:39.91
[root@mlb-dc1-centos7 ~]#
[root@mlb-dc1-centos7 ~]# ucdDemoUserList.0
-bash: ucdDemoUserList.0: command not found
[root@mlb-dc1-centos7 ~]# snmpget -v 2c -c public 10.0.1.2 sysUpTime.0 ucdDemoUserList.0
DISMAN-EVENT-MIB::sysUpTimeInstance = Timeticks: (252132) 0:42:01.32
UCD-DEMO-MIB::ucdDemoUserList.0 = No Such Object available on this agent at this OID
[root@mlb-dc1-centos7 ~]# _
```

snmpgetnext: retrieving unknown indexed data.

